a) **Silicone rubbers:** These silicones are bridged together by methylene or similar groups

b) **Silicone resins:** They are obtained by blending silicones with organic resins such as acrylic esters.

(ii) Cyclic silicones

These are obtained by the hydrolysis of R_2SiCl_2.

(iii) Cross linked silicones

They are obtained by hydrolysis of $RSiCl_3$

Properties

The extent of cross linking and nature of alkyl group determine the nature of polymer. They range from oily liquids to rubber like solids. All silicones are water repellent. This property arises due to the presence of organic side groups that surrounds the silicon which makes the molecule looks like an alkane. They are also thermal and electrical insulators. Chemically they are inert. Lower silicones are oily liquids whereas higher silicones with long chain structure are waxy solids. The viscosity of silicon oil remains constant and doesn't change with temperature and they don't thicken during winter

Uses:

1. Silicones are used for low temperature lubrication and in vacuum pumps, high temperature oil baths etc...
2. They are used for making water proofing clothes
3. They are used as insulating material in electrical motor and other appliances
4. They are mixed with paints and enamels to make them resistant towards high temperature, sunlight, dampness and chemicals.

2.3.9 Silicates

The mineral which contains silicon and oxygen in tetrahedral $[SiO_4]^{4-}$ units linked together in different patterns are called silicates. Nearly 95 % of the earth crust is composed of silicate minerals and silica. The glass and ceramic industries are based on the chemistry silicates.

Types of Silicates:

Silicates are classified into various types based on the way in which the tetrahedral units, $[SiO_4]^{4-}$ are linked together.

Ortho silicates (Neso silicates): The simplest silicates which contain discrete $[SiO_4]^{4-}$ tetrahedral units are called ortho silicates or neso silicates.

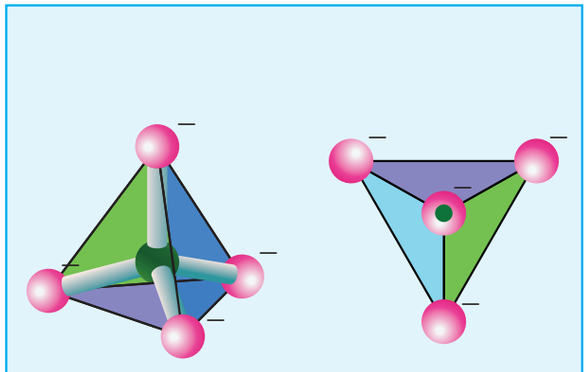

Figure 2.11
Structure of Ortho silicates

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